

Apple HomeKit. HomeKit is a home automation platform that lets you control all compatible home appliances securely and conveniently using an iOS device. The idea is to be able to install several disparate devices in your house, forget all about their individual apps and manage these using HomeKit.

You can pre-program single commands or moods that achieve several tasks at one go. For example, when you tell the app that you are going to bed, it automatically dims the lights, adjusts the temperature of your bedroom's air-conditioner, turns on the bedside reading lights and so on. With Siri, HomeKit is also voice-enabled.

Windows 10 and Cortana. Microsoft hopes to achieve a similar feat in the near future by placing Windows 10 and Cortana at the realm of smarthomes. It will enable the same kind of unified access to devices with voice and gesture support, thanks to Cortana. Although currently Windows is compatible only with devices running AllJoyn protocol, it soon plans to work with a wide range of devices and appliances by integrating the new Open Connectivity Foundation (OCF) protocols, scheduled to be released in 2017.

Amazon Echo. Amazon's Echo, voice-enabled with Alexa, has already made a mark for itself in the home automation space. The new, low-cost version called Echo Dot, also voice-enabled, seems equally successful. Wake up your Echo by calling it Amazon, Echo or Alexa, give it an instruction or ask a question and experience the convenience of having a personal assistant.

Echo can play music, read books, tell you the weather and control smart home devices from multiple makers like WeMo, Philips Hue, Samsung SmartThings, Wink, Insteon, Nest and ecobee.

Intraix KLUG. A recent innovation, KLUG Home from Intraix, is packaged like a simple universal serial bus (USB) stick that plugs

into your Wi-Fi router. It connects wirelessly to a variety of smart devices from various manufacturers and uses machine learning to study your usage patterns and learn your personal preferences. This helps it to automatically manage your home. You can also manually voice-control the devices connected to KLUG.

Inoho. In India, a Bangalore based startup called Inoho is quietly growing, with a home automation solution that is perfect for an average



ecobee3 Smarter Wi-Fi thermostat
(Image courtesy: www.ecobee.com)

Smart switches to connect any device

- ▶ Global bestseller: Belkin WeMo Insight Switch
- ▶ Made-in-India: Pert, Inoho, eGlu, tagPlug and Xenon SmartPlug

Indian household, where existing devices such as tubelights, fans and geysers might not really be smart. Inoho adds smartness to dumb devices, too, and enables you to manage all of these using a single app.

"Managing multiple connected devices, when each device is connected individually, can be a pain in the neck. It becomes a connected chaos. Connected devices in a home can have different manifestations or functions, for example, on and off for lights, drawing and raising of curtains, video feeds from cameras, cooking functions on the microwave, wash cycles in a washing machine and so on.

"Eighty per cent of the interac-

tions in a home are on/off interactions, which is what our first product targeted. We have now integrated other devices as well, for example, motorised curtain controls and connected video cameras," says Deepankar Garg, co-founder of Inoho.

At the heart of Inoho solution is the home controller, a central platform that talks to the peripherals connected to it. It consists of a processing unit and communication modules. Their first peripheral is a switchboard module, which fits easily behind the switchboards at home. It combines communications, processing and actuators (to achieve on and off).

Then, there is the Cloud server that can help users access, control and monitor devices from anywhere in the world using the app as the front-end. There is a Web version of the app that can be run on any browser, as well as mobile versions for iOS and Android devices.

"Inoho has not covered the universe of functions or devices, for that matter, I do not think anyone has. Integration onto the platform is going to be a continuous process. And we will be continuously adding more products and functions and tying up with appliance manufacturers so that users can do everything centrally, and all devices can learn from and talk to each other. It is a continuous process," explains Garg.

In the days to come, Inoho plans to enhance their switchboard module with different configurations and form factors, add more peripherals to their platform and also enhance the platform itself so that functionality and intelligence of the platform is more than the sum of its individual parts.

Adding security to the equation

With constant developments in embedded hardware and operating OSes, displays and user interfaces, it is possible to develop a smart product for almost any requirement. And if you make it compatible with one